Predicting Customer Churn in a Telecommunications Dataset

Charles Sarratt

DSC630-T302 Predictive Analytics (2263-1)

Professor Andrew Hua

March 8, 2026

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Introduction

Customer churn represents a significant challenge for subscription-based businesses, particularly within the telecommunications industry. When customers discontinue service, organizations experience immediate revenue loss, increased customer acquisition costs, and potential long-term brand impact. Because retaining existing customers is generally more cost-effective than acquiring new ones, the ability to accurately identify customers at risk of churning has substantial financial and strategic value.

This project focuses on predicting customer churn using the Telco Customer Churn dataset. The primary objective is to develop and evaluate classification models capable of identifying customers who are likely to leave service. In addition to predictive performance, this analysis seeks to determine which customer characteristics are most strongly associated with churn behavior. By understanding the drivers of attrition, organizations can design targeted retention strategies, optimize contract structures, and improve customer engagement efforts.

The intended audience for this analysis includes business leaders, data analysts, marketing teams, and customer success managers responsible for retention strategy and revenue stability. Insights generated from predictive modeling can support proactive decision-making, allowing organizations to intervene before customers cancel service. Ultimately, this project aims to demonstrate how structured customer data and classification modeling techniques can be leveraged to produce actionable business intelligence.

Data

The analysis utilizes the Telco Customer Churn dataset, a publicly available dataset hosted on Kaggle. The dataset contains structured information on customer demographics, service subscriptions, contract types, payment methods, tenure, and monthly charges, along with a binary churn indicator identifying whether a customer discontinued service. The dataset originally contained over 7,000 customer records and features commonly associated with churn behavior in subscription-based industries.

The target variable, *Churn*, indicates whether a customer left the company (1) or remained active (0). Because churn prediction is a binary classification problem, this variable serves as the dependent variable in the modeling process. The dataset includes a mixture of numerical and categorical predictors, requiring preprocessing prior to model implementation.

During data preparation, the *TotalCharges* variable was converted from object format to numeric format, and invalid or blank entries were treated as missing values. Rows containing missing values for this feature were removed to preserve data integrity. The *customerID* variable was excluded from modeling because it does not provide predictive value. Categorical variables were transformed using one-hot encoding to ensure compatibility with classification algorithms, and numerical features were standardized using z-score scaling to improve model stability.

After preprocessing, the final modeling dataset contained 7,032 customer records. The data were divided into training and testing sets using an 80/20 stratified split to preserve the original class distribution. Approximately 5,625 observations were allocated to the training set and 1,407 to the testing set. The churn rate in the dataset is approximately 26 percent, indicating moderate class imbalance and reinforcing the importance of evaluation metrics such as recall and F1 score during model assessment.

Methods of Analysis

To address the research objective of predicting customer churn, multiple classification models were evaluated. Logistic regression was implemented as a baseline classifier due to its interpretability and suitability for binary classification problems. Because understanding the drivers of churn is as important as predictive performance, logistic regression provides the added benefit of coefficient interpretation while maintaining strong generalization capability.

The logistic regression model was implemented with L2 regularization to reduce the risk of overfitting and improve stability on unseen data. Prior to modeling, all predictor variables were standardized using z-score scaling to ensure that coefficients were comparable in magnitude and that the model converged efficiently.

To evaluate whether nonlinear modeling techniques would improve predictive performance, a Random Forest classifier was also implemented using the same preprocessing pipeline and identical train-test split. This allowed for direct comparison between a linear model and a nonlinear ensemble method.

Model performance was evaluated using metrics appropriate for binary classification under moderate class imbalance. Although overall accuracy was reported, greater emphasis was placed on precision, recall, and F1 score for the churn class. Because churn represents approximately 26 percent of the dataset, accuracy alone could mask poor minority-class performance. Recall is particularly important in churn prediction, as false negatives represent customers who leave without being identified in advance. The receiver operating characteristic (ROC) curve and area under the curve (AUC) were also used to evaluate the models' ability to distinguish between churn and non-churn customers across varying classification thresholds.

All models were trained using the 80 percent training dataset and evaluated on the 20 percent test dataset to assess generalization performance.

Results and Findings

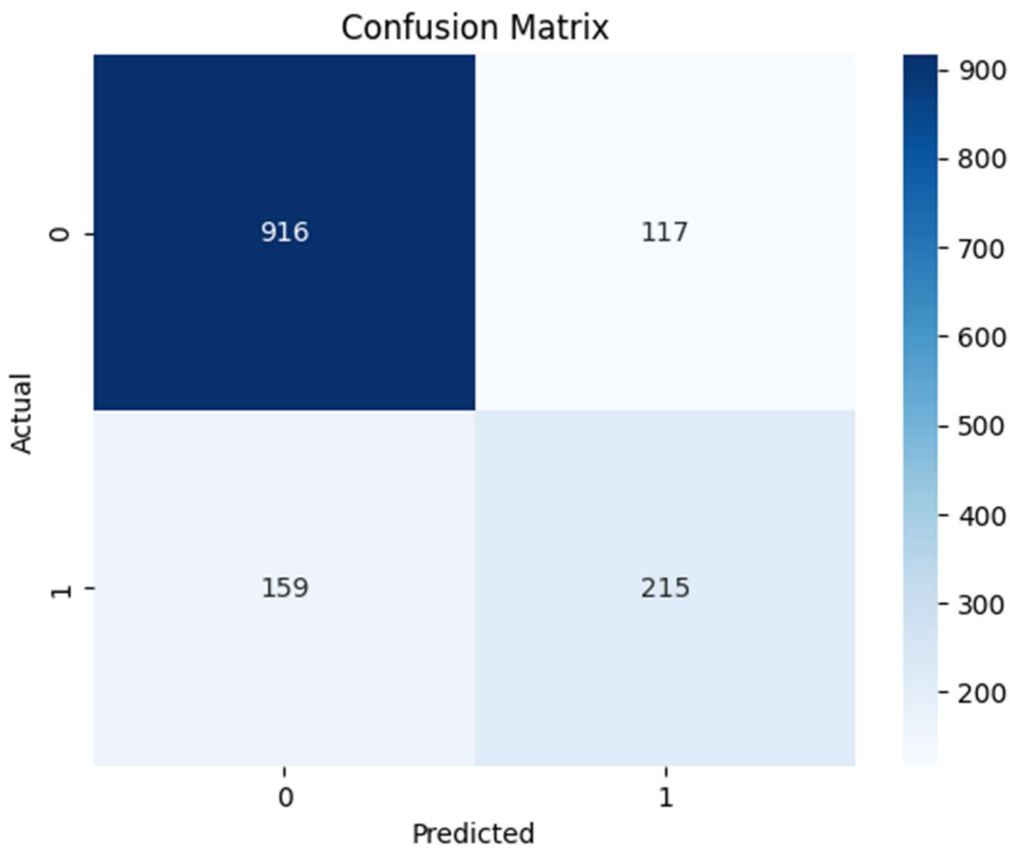
The logistic regression model achieved an accuracy of 80.4 percent on the test dataset. Precision for the churn class was 0.65, recall was 0.57, and the resulting F1 score was 0.61. The ROC-AUC value of 0.84 indicates strong discriminative ability, demonstrating that the model effectively distinguishes between churn and non-churn customers across varying probability thresholds.

The classification report confirms stronger performance in predicting non-churn customers compared to churn customers, which is expected given the class imbalance. However, performance metrics for the churn class remain meaningful and actionable, particularly in the context of business decision-making.

As shown in Figure 1, the confusion matrix provides further insight into prediction outcomes.

Figure 1

Confusion Matrix for Logistic Regression Model



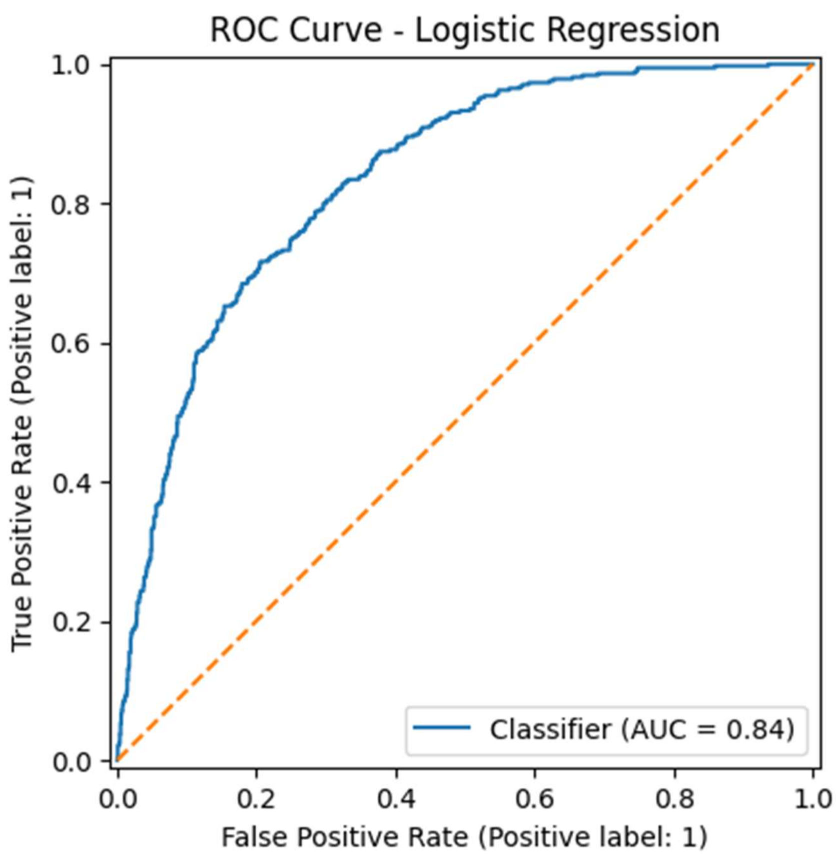
Note. The matrix displays counts of true positives, true negatives, false positives, and false negatives for churn classification.

The model correctly identified 916 non-churn customers and 215 churn customers. However, 159 churn cases were misclassified as non-churn. The relatively high number of false negatives compared to true positives explains the moderate recall value and highlights the tradeoff between identifying churners and avoiding unnecessary false alarms. Because false negatives represent customers who churned without intervention, improving recall would directly increase the number of at-risk customers identified for retention efforts.

As shown in Figure 2, the ROC curve remains above the no-skill reference line across classification thresholds, and the area under the curve of 0.84 confirms strong classification performance. This indicates that the model maintains effective separation between churn and non-churn customers across classification thresholds.

Figure 2

ROC Curve for Logistic Regression Model



Note. The ROC curve illustrates model discrimination across classification thresholds (AUC = 0.84).

To determine whether nonlinear modeling techniques would improve performance, a Random Forest classifier was evaluated using the same training and testing framework. The Random Forest model achieved an accuracy of 78.9 percent, precision of 0.63, recall of 0.52, F1 score of 0.57, and ROC-AUC value of 0.82. Although Random Forest is capable of capturing nonlinear relationships, it did not outperform logistic regression across key evaluation metrics. Logistic regression demonstrated stronger recall, F1 score, and overall AUC performance, leading to its selection as the preferred model.

Interpretation of the logistic regression coefficients reveals several meaningful predictors of churn. Tenure shows a strong negative relationship with churn, indicating that customers with longer service duration are less likely to leave. Contract type is highly influential, as customers on one-year and two-year contracts are significantly less likely to churn compared to those on month-to-month agreements. Monthly charges and fiber optic internet service are positively associated with churn, suggesting that higher-cost or premium service plans may increase attrition risk. Payment method, particularly electronic check, also exhibits a positive association with churn behavior.

These relationships align with well-documented churn patterns in subscription-based industries. The consistency between statistical findings and realistic customer behavior strengthens confidence in both the predictive validity and practical relevance of the model. The fact that logistic regression outperformed the more complex Random Forest model suggests that churn behavior in this dataset is influenced by relatively structured and linearly separable relationships among variables. This supports prioritizing interpretability while maintaining strong predictive performance.

Conclusion

This analysis demonstrates that customer churn can be predicted with strong discriminative performance using structured customer data and classification modeling techniques. The logistic regression model achieved solid overall accuracy while maintaining meaningful recall and F1 score for the churn class. The ROC-AUC value of 0.84 further confirms the model's ability to distinguish between customers who are likely to churn and those who are not.

Several customer characteristics were identified as influential predictors of churn, including tenure, contract type, monthly charges, internet service type, and payment method. Customers with shorter tenure and month-to-month contracts exhibit elevated churn risk, while longer contract commitments and extended service duration are associated with increased retention. These findings align with established industry patterns and provide actionable insight for business decision-making.

From a practical perspective, organizations may consider prioritizing retention efforts for customers with short tenure, flexible contract structures, and higher monthly charges. Early engagement strategies, contract incentives, and proactive outreach programs may reduce attrition among high-risk segments. However, predictive performance should be evaluated alongside operational costs and retention program effectiveness to ensure that interventions generate positive return on investment.

Although the model demonstrates strong performance, limitations should be acknowledged. The presence of false negatives indicates that some at-risk customers remain undetected. Additionally, the analysis is based on historical data from a single telecommunications dataset, which may limit generalizability to other industries or time periods. Future work may explore threshold adjustment, class weighting techniques, or additional ensemble methods to further improve recall while maintaining overall model stability.

Overall, this study illustrates the value of predictive analytics in identifying churn risk and supporting data-driven retention strategies. Importantly, the results also demonstrate that increased model complexity does not automatically yield superior performance. In this case, the simpler and more interpretable logistic regression model outperformed a more complex ensemble method, reinforcing the importance of thoughtful model evaluation and alignment with business objectives.

References

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